Lecture 2

From Sample to Sequence

Day 1 · 09:45–10:30 · 45 min

LO2: describe Illumina and Nanopore outputs and choose between them · LO6 (part): understand the outsourcing workflow

The journey of an isolate

Wound swab → **Culture** → **Pure isolate**

DNA extraction → **Library prep** → **Sequencing** → **Data files** → **Analysis**

KEY MESSAGE

Everything up to the pure isolate (dark boxes) already happens in our labs.

We sequence isolates, not swabs

- An **isolate** is a pure culture of one bacterium, so its genome is clean
- Sequencing a swab directly (**metagenomics**) mixes human DNA with many microbes: harder and more expensive
- For SSI surveillance, **isolate sequencing is the standard**

KEY MESSAGE

Store your isolates (–80 °C or glycerol stocks). You can't sequence what you threw away.

DNA extraction and library preparation

- **Extraction:** break open the cells and purify the DNA
- **Library prep:** cut or tag the DNA and add short “adapter” sequences so the machine can read it
- **Multiplexing:** many samples are barcoded and run together, which lowers the cost per genome

> You don’t need to do this yourself. But poor-quality DNA gives poor-quality data (Lab 2).

Two main technologies

	Illumina (short reads)	Oxford Nanopore (long reads)
Read length	100–300 letters, in pairs	thousands to 100,000+ letters
Accuracy	very high	now high with modern chemistry
Best for	SNP-level outbreak work	complete genomes, **plasmids**
Machine	large, central labs	MinION is pocket-sized
Files	FASTQ pair (R1 + R2)	one FASTQ

Jigsaw analogy: Illumina gives many tiny pieces; Nanopore gives fewer, bigger pieces.

What is a read?

- A **read** is one piece of DNA sequence produced by the machine
- One bacterial genome gives **hundreds of thousands to millions** of reads
- We put the reads back together into the genome

> Like rebuilding a shredded book from many overlapping copies.

Coverage (depth)

Coverage = how many reads, on average, cover each position

$$(\text{number of reads} \times \text{read length}) \div \text{genome size}$$

$$1{,}000{,}000 \times 150 \div 5{,}000{,}000 = \mathbf{30\times}$$

KEY MESSAGE

Aim for ≥ 30× with Illumina and ≥ 40–50× with Nanopore.

Which should I use?

Question	Best choice
Is this an outbreak? (fine SNP differences)	Illumina
Is resistance on a plasmid moving between species?	Nanopore or hybrid
Need results fast, locally	Nanopore
Lowest cost per genome at large scale	Illumina (batched)
Best of both	**Hybrid** assembly

KEY MESSAGE

Your sequencing partner may use either. This course teaches both.

Outsourcing: how it works

Agree on a partner → **Ship isolates or DNA** → **Partner sequences** → **You analyse**

- **Partners:** national reference lab, Africa CDC network, academic collaborator, or a commercial service
- **Shipping:** follow their biosafety rules; you may need permits and a material transfer agreement
- **You get back:** FASTQ files, sometimes assemblies and reports. **Keep the data.**

Questions to ask a sequencing partner

Platform?
Illumina, Nanopore, or both

Coverage?
≥ 30× Illumina

Raw reads?
Make sure FASTQ files are included

Turnaround?
Weeks from shipping to data

Cost?
Per isolate, including shipping

Data sharing?
Who owns and publishes the data

Metadata: the most underrated part

A genome without **when, where and from whom** is almost useless for surveillance.

Minimum fields

- isolate ID · collection date
- facility · ward · specimen type
- surgery type and date
- AST results

PROTECT PRIVACY

Never put names or patient IDs in shared files. Use coded study IDs.

Standard: **PHA4GE** contextual data specification

Which sample, from where?

Field	Example for one SSI isolate
specimen collector sample ID	`UGHE-SSI-0001` (your lab's own ID)
specimen source ID	`CASE-0042`: coded, never a name
geo_loc name (country)	Rwanda
geo_loc name (state/province/region)	Northern Province

Who handled it?

Field	Example for one SSI isolate
sample collected by	Full official name of the hospital lab
sequenced by	Full official name of the sequencing partner
sequence submitted by	Who uploaded it to ENA / NCBI
data custodian organization name	Who decides how the data are used

Use full official names, not abbreviations.

When, what and why?

Field	Example for one SSI isolate
sample collection date	`2026-02-09` (YYYY-MM-DD, to the day)
organism	*Staphylococcus aureus*
purpose of sampling	Diagnostic testing
purpose of sequencing	Cluster/Outbreak investigation

Add your SSI fields

The 12 PHA4GE fields are the **minimum** for any pathogen. For SSI surveillance, add:

- ward · specimen type (e.g. wound swab)
- surgery type · surgery date · theatre
- AST results

KEY MESSAGE

Same column names in every hospital = data you can combine.

What files come back?

File	What it is
`sample_R1.fastq.gz` + `sample_R2.fastq.gz`	Illumina reads (a pair)
sample.fastq.gz	Nanopore reads
sample.fasta	Assembly (sometimes)
QC report	Sometimes

> In Lab 1 we'll open these files and look inside.

Quick check (show of hands)

1. You want to know if **6 MRSA isolates from one ward** are an outbreak. Which data would you request?

2. You suspect a **carbapenemase gene is jumping** between *Klebsiella* and *E. coli*. Which data would you request?

Quick check: answers

1. **Illumina, ≥ 30×**, with collection dates and ward metadata

2. **Nanopore or hybrid**, to resolve the plasmids